IRVAN TRIANTO

(+62) 823-8955-8743 | irvntriantoo@gmail.com | LinkedIn | Portofolio

PROFESSIONAL SUMMARY

Geospatial professional with hands on experience in spatial data management, digital mapping, and GIS based project support within a government/BUMN infrastructure project (PLN floating solar power plant, 100 MWac). Skilled in processing spatial data in GeoJSON, SHP, KML, and Excel/CSV formats, working with GPS coordinates and projection systems, and managing spatial databases (PostgreSQL/PostGIS, basic). Familiar with Leaflet for WebGIS visualization. Brings distinct domain strength in marine and coastal/fisheries related geospatial data from hydrographic survey and coral reef conservation project experience. Comfortable coordinating between technical and operational teams, with a track record of cross disciplinary collaboration in field and research settings.

RELEVANT EXPERIENCE

PT Gema Sarana Teknik | Batam, Indonesia Aug 2025 – Aug 2026

Surveyor — Navigation, Positioning & Geospatial Data Processing

- Conducted geospatial and hydrographic data collection and digital mapping across multiple survey projects over a 12 month engagement, including a 3 month deployment on the Karangkates Floating Photovoltaic Solar Power Plant (100 MWac), a PLN (state owned) infrastructure project.
- Processed satellite imagery and produced thematic spatial outputs, including nautical charts, mangrove distribution maps, and seagrass (lamun) mapping, using GIS software to integrate field survey data with spatial datasets.
- Operated GNSS/GPS positioning systems and hydrographic survey instruments, working directly with GPS coordinates and projection systems in offshore and coastal environments.
- Performed spatial data validation and cross verification between field observations, satellite imagery, and CAD/survey records to ensure accuracy and consistency of geospatial deliverables.
- Coordinated with cross functional field teams under project specifications, comparable to liaising between technical and operational stakeholders.

PT Amati Karya Indonesia | Jakarta, Indonesia Feb 2024 – Jun 2024

Project Coordinator — Coral Reef Conservation Team

- Processed and interpreted satellite imagery to support environmental monitoring, spatial analysis, and coastal ecosystem mapping for conservation planning.
- Compiled, analyzed, and visualized spatial data into thematic maps supporting field based project decision making.
- Coordinated with a multidisciplinary team (researchers, field staff, community stakeholders) to integrate GIS based environmental assessment throughout the project cycle.
- Presented spatial analysis results and project outcomes at an international forum.

Universitas Negeri Semarang | Semarang, Indonesia Sep 2024 – Sep 2025

Lab Supervisor

- Assisted Remote Sensing practicum sessions, guiding students in satellite imagery processing, image interpretation, and geospatial analysis using GIS and remote sensing software.
- Provided technical support for GIS and Remote Sensing laboratory activities, including data preparation, software troubleshooting, and spatial data processing workflows.
- Verified and maintained the accuracy of 100+ academic records, achieving 100% data compliance through systematic quality control procedures.
- Oversaw laboratory operations for one full year, ensuring compliance with laboratory procedures and safety standards during practical sessions.

Universitas Negeri Semarang | Semarang, Indonesia Jul – Aug 2025

Administrative Assistant

- Supported administrative operations and documentation within the academic department.

FREELANCE & PROJECT EXPERIENCE

Measuring Current Project | Batam, Indonesia Feb 2026

Hydrology Surveyor

- Conducted offshore environmental field surveys under dynamic sea conditions using ADCP and Current Meter instruments, following operational safety procedures throughout data collection.
- Ensured quality assurance and validation of hydrodynamic data to support environmental and engineering risk assessment.

Community Service Program | Karanganyar, Indonesia Dec 2024 – Jan 2025

Project Coordinator

- Led 15 community development projects, integrating social and environmental safety considerations into field activity planning.
- Supervised Trichoderma sp. cultivation training at the tissue culture laboratory of Tahura KGPAA Mangkunagoro I, reinforcing safe handling of laboratory materials.
- Conducted hygiene and safety inspections of production kitchen facilities in partnership with Gapoktan (farmer group association).

Study of Geosphere Phenomena 2 | Karimunjawa Island, Indonesia Feb – Jun 2023

Project Coordinator

- Led a high-risk nearshore environmental survey assessing coastal abrasion patterns and shoreline change at Bobby Beach, Karimunjawa Island, contributing spatial and field-based evidence for environmental impact assessment.
- Implemented life jacket protocols and weather/wave checks prior to fieldwork, overseeing team safety in a disturbed coastal environment.

EDUCATION

Universitas Negeri Semarang — Bachelor's Degree in Geography, GPA 3.79/4.00

Semarang, Indonesia | 2021 – 2025

- 2022 – Top 10 Finalist: Faculty Ambassador (Duta Fakultas), FISIP, Universitas Negeri Semarang.

CERTIFICATIONS

- Ahli K3 Umum (AK3U) Training, 2026 — PT Sarana Inspirasi Maju Bersaudara (Professional Project Institute), Batam. Official Kemnaker certification currently in the issuance process.
- International Certificate in Sustainable Marine Ecosystems, 2024 — Ghent University & ASEAN South Asia Platform.
- TOPEL (Score: 443), 2025 — Institutional Testing Program.
- Internship Certification – IS-CORE Wallacea Project, 2024 — PT Amati Karya Indonesia, in collaboration with ASEAN South Asia Platform.

SCIENTIFIC PUBLICATIONS

- Connectivity Index Analysis of Religious Tourism Sites in Semarang City to Support Tourist Visits. GeoImage Journal, Vol. 15 No. 1 (2026). https://doi.org/10.15294/geoimage.v15i1.40537
- Trichoderma sp. Cultivation Training as a Natural Fungicide, Green Lawu Forest Farmer Group, Karanganyar. ABDIMAS: Jurnal Pengabdian Masyarakat, Vol. 29 No. 1 (2025). https://doi.org/10.15294/q7khw835
- The Effect of Land Cover on the Urban Index of Semarang City. Proceedings of the FISIP UNNES National Seminar (2024). https://proceeding.unnes.ac.id/psnf/article/view/2913

SKILLS

- GIS Software & Formats: QGIS (proficient), ArcGIS, ArcGIS Pro, GeoJSON, SHP, KML, spatial data validation and QA/QC.
- Coordinate Systems & Positioning: GPS/GNSS positioning, latitude longitude coordinates, projection systems, field based coordinate verification.
- Database & WebGIS: PostgreSQL/PostGIS (basic), Leaflet (basic), spatial database management, spatial and non spatial data structuring and integration.
- Data Processing: Excel/CSV spatial data handling, large volume data processing, secondary data processing.
- Domain Expertise: Marine, coastal, and fisheries related geospatial data (hydrographic survey, coral reef and coastal ecosystem mapping); government/BUMN infrastructure project experience (PLN floating solar power plant).
- Collaboration & Communication: Liaison between technical and operational teams; cross disciplinary teamwork with field, research, and stakeholder groups; report writing in Bahasa Indonesia and English.